

ARJUN GIRIRAJ DAY-04

1. What is a network protocol?

A set of rules which is followed by device for communicating.

2. What is the OSI Model?

Open System Interconnection is a 7 layer model to define how system communicate over network.

3. Name all seven OSI layers in correct order.

- 1) Physical
- 2) Data Link
- 3) Network
- 4) Transport
- 5) Session
- 6) Presentation
- 7) Application

4. What is the purpose of the Application Layer?

Provides Network service to Application Using HTTP, HTTPS, SMTP, FTP, DNS, etc.

5. What is the purpose of the Transport Layer?

Helps in Transportation of data using TCP/UDP.

6. What is the purpose of the Network Layer?

Addressing the devices IPV4,IPV6 and Routing.

7. What is the purpose of the Data Link Layer?

Local network communication and physical addressing(MAC).

8. What is the purpose of the Physical Layer?

It transmits bit streams over physical media like cable, fiber.

9. What is the TCP/IP Model?

It is the simpler and practical version of OSI models, it consists of 4 layers instead of 7 layers in OSI models.

10. Explain the difference between OSI and TCP/IP models.

OSI model is a theoretical model consisting of 7 layers, and the TCP/IP model is a practical model consisting of 4 layers.

11. What is encapsulation?

It is the process in computer network where data moves down the protocol stack and each layer adds its header.

12. What is decapsulation?

It is the process in computer network where data received at the receiver end and moves up the protocol stack.

13. Which layer uses IP addresses?

Network Layer.

14. Which layer uses MAC addresses?

Data Link Layer.

15. Which layer uses TCP and UDP?

Transport Layer.

16. Which layer includes HTTP and HTTPS?

Application Layer.

17. Why should cybersecurity professionals understand networking layers?

Cybersecurity analysts investigate these networking layers. And this is the foundation for Wireshark, Nmap, BurpSuite, and Pentesting.

18. Explain how network data travels from a browser to a server.

The browser resolves the domain by DNS, then it opens the tcp connection to send the data on packets, Packets are routed to the server and server replies with HTTP response.

19. Your resolved IP address for google.com (nslookup).

Name: google.com

Address: 192.178.158.102

Name: google.com

Address: 192.178.158.113

Name: google.com

Address: 192.178.158.139

Name: google.com
Address: 192.178.158.138
Name: google.com
Address: 192.178.158.100
Name: google.com
Address: 192.178.158.101
Name: google.com
Address: 2404:6800:4013:804::65
Name: google.com
Address: 2404:6800:4013:804::66
Name: google.com
Address: 2404:6800:4013:804::71
Name: google.com
Address: 2404:6800:4013:804::8a

20. Number of traceroute hops to [google.com](https://www.google.com).
30 hops

ping [google.com](https://www.google.com)

```
venom@VENOM:~$ ping google.com
PING google.com (2404:6800:4013:804::8b) 56 data bytes
64 bytes from nw-in-f139.lel100.net (2404:6800:4013:804::8b): icmp_seq=1 ttl=111
time=606 ms
64 bytes from nw-in-f139.lel100.net (2404:6800:4013:804::8b): icmp_seq=2 ttl=111
time=426 ms
64 bytes from nw-in-f139.lel100.net (2404:6800:4013:804::8b): icmp_seq=3 ttl=111
time=659 ms
64 bytes from nw-in-f139.lel100.net (2404:6800:4013:804::8b): icmp_seq=4 ttl=111
time=33.1 ms
64 bytes from nw-in-f139.lel100.net (2404:6800:4013:804::8b): icmp_seq=5 ttl=111
time=39.3 ms
64 bytes from nw-in-f139.lel100.net (2404:6800:4013:804::8b): icmp_seq=6 ttl=111
time=47.0 ms
64 bytes from nw-in-f139.lel100.net (2404:6800:4013:804::8b): icmp_seq=7 ttl=111
time=340 ms
64 bytes from nw-in-f139.lel100.net (2404:6800:4013:804::8b): icmp_seq=8 ttl=111
time=170 ms
64 bytes from nw-in-f139.lel100.net (2404:6800:4013:804::8b): icmp_seq=9 ttl=111
time=399 ms
64 bytes from nw-in-f139.lel100.net (2404:6800:4013:804::8b): icmp_seq=10 ttl=111
time=351 ms
64 bytes from nw-in-f139.lel100.net (2404:6800:4013:804::8b): icmp_seq=11 ttl=111
time=314 ms
```

tracert [google.com](https://www.google.com)

```
venom@VENOM~$ traceroute google.com
traceroute to google.com (2404:6800:4013:804::8a), 30 hops max, 60 byte packets
 1  2401:4900:a5f7:e636::fc (2401:4900:a5f7:e636::fc)  35.931 ms  35.898 ms  35.882 ms
 2  * * *
 3  fc00:0:2:231::1 (fc00:0:2:231::1)  35.065 ms * *
 4  * * *
 5  2404:a800:1a00:805::19 (2404:a800:1a00:805::19)  29.670 ms  29.615 ms  29.507 ms
 6  * * *
 7  * * 2001:4860:1:1::14d4 (2001:4860:1:1::14d4)  23.294 ms
 8  2001:4860:0:1::78d3 (2001:4860:0:1::78d3)  23.228 ms 2001:4860:0:1::7819 (2001:4860:0:1::7819)  23.198 ms  94.455 ms
 9  2001:4860:0:1::77dc (2001:4860:0:1::77dc)  99.029 ms  98.998 ms 2001:4860:0:1::78c0 (2001:4860:0:1::78c0)  139.430 ms
10  2001:4860:0:1::779f (2001:4860:0:1::779f)  125.512 ms 2001:4860:0:1::77c7 (2001:4860:0:1::77c7)  125.733 ms 2001:4860:0:1::7841 (2001:4860:0:1::7841)  82.896 ms
11  2001:4860:0:1::7e2c (2001:4860:0:1::7e2c)  82.829 ms 2001:4860:0:1::a164 (2001:4860:0:1::a164)  59.297 ms 2001:4860:0:1::7e2c (2001:4860:0:1::7e2c)  59.259 ms
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
```

nslookup [google.com](https://www.google.com)

```
venom@VENOM~$ nslookup google.com
Server:          10.242.36.44
Address:         10.242.36.44#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.158.102
Name:   google.com
Address: 192.178.158.113
Name:   google.com
Address: 192.178.158.139
Name:   google.com
Address: 192.178.158.138
Name:   google.com
Address: 192.178.158.100
Name:   google.com
Address: 192.178.158.101
Name:   google.com
Address: 2404:6800:4013:804::65
Name:   google.com
Address: 2404:6800:4013:804::66
Name:   google.com
Address: 2404:6800:4013:804::71
Name:   google.com
Address: 2404:6800:4013:804::8a
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